Practice Exam - 1

Question 65 of 117

A 72-year-old male with longstanding history of obesity (BMI 42 Kg/m2), DM, and HTN presents to clinic for evaluation for worsening DOE and abdominal bloating. On examination, he has no LE edema or ascites. JVP is estimated at 6 mm of H20. EKG reveals NSR nonspecific IVCD. CXR is unremarkable. TTE revealed a LVEF of 50%, normal RV function, PASP of 48 mmHg, grade 3 diastolic dysfunction with an E/e' of 16. Blood work is notable for Na 137 mmol/L, K 4.5 mEq/L, BUN 28 mg/dl, Cr 1.4 mg/dl. CBC is unremarkable. NT proBNP is 432 pg/ml. TSH is within normal limits.[MRC1.1]

He is diagnosed with HFpEF and initiated on empagliflozin 10 mg daily. He feels better but continues to report dyspnea and abdominal bloating. NT proBNP remains elevated. Stress testing performed with PET CT was negative for ischemia. Vital remains stable with a BP of 128mg/83 mmHg , HR 84 bpm.

Which of the following should be the next step in his treatment

A. Initiate spironolactone 25 mg

✓ **B.** Initiate furosemide 20mg daily

C. Initiate entresto 24-26mg twice daily

D. Initiate isosorbide mononitrate 30 mg daily

Commentary References

Content code of Blueprint: HFpEF

Task: Treatment

Teaching Point: Exam takers should know the data supporting the role of various therapies in HFpEF

Rationale:

The patient has a diagnosis of HFpEF with some improvement with initiation of empagliflozin however he remains volume overloaded based on symptoms and natruretic peptide's. As such, the appropriate neck step would be decongestion with loop diureti (Choice B).

Initiating the patient on spironolactone 25 mg (Choice A) would be reasonable after initiation of loop diuretics however would not be the correct next step nor is the subsequent reasoning of choice a correct. The TOP CAT trial of spironolactone in HFpEF did not show statistically significant improvement in the composite endpoint of death, aborted cardiac catheter HF hospitalization between the spironolactone and placebo groups. However, it is imperative to note that post hoc analyses showed differential effects in the Americas as compared to Russia–Georgia with the potential benefits seen in the Americas (LVEF ≥45%, elevated BNP level or HF admission within 1 year, eGFR >30 mL/min/1.73 m2, creatinine <2.5 mg/dL, and potassium <5.0 mEq/L). A separate postop analysis also showed that EF also showed an interaction with effective spironolactone with the highest effect seen with LVEF less than 50%.

The initiation of an ARNi is also reasonable and HFpEF patient's however the Paragon–HF trial did not achieve a statistically significant difference in its primary composite outcome of cardiovascular death or HF hospitalization. Exploratory post hoc analyses did show that Entresto appeared to have a greater effect in patients with an EF <57% and in women more than men.

Lastly,NEAT-HFpEF (Nitrate’s Effect on Activity Tolerance in Heart Failure With Preserved Ejection Fraction) trial evaluating isosorbide mononitrate use and HFpEF patient's did not show any benefit across all outcomes.